FBISE

INTERMEDIATE (11TH CLASS) EXAMINATION, 2022

COMPUTER SCIENCE

Past Paper 2022 • Paper Code: 1HA-I 2212 — D(7121)

Retyped faithfully from the original board paper — same questions, no changes.

Note: COMPUTER SCIENCE HSSC-I. SECTION – A (Marks 15). Time allowed: 20 Minutes. Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed. Do not use lead pencil. Version No. 7121. Fill the relevant bubble against each question:

Q1. Which of the following is an example of embedded computing device?

(A) Keyboard
(B) Digital Washing Machine
(C) Mouse
(D) Trackball

Q2. Which of the following is used to process complex calculations and designing of complicated machines?

(A) Mini computer
(B) Micro computer
(C) Personal computer
(D) Super computer

Q3. Which of the following types of optical disks can store more than 100 GB of data?

(A) Blu-ray disk
(B) Digital video disk
(C) Compact disk
(D) Floppy disk

Q4. A type of memory that needs frequent power refreshing to retain its charge is called:

(A) Programmable ROM
(B) Magneto resistive RAM
(C) Dynamic RAM
(D) Static RAM

Q5. The part of computer system that can hold data temporarily during processing is called:

(A) Register
(B) ROM
(C) Optical disk
(D) Hard disk

Q6. Which of the following registers holds program instruction that are fetched from the memory for execution?

(A) Memory Address Register
(B) Memory Buffer Register
(C) Program Counter Register
(D) Instruction Register

Q7. The expansion card which is required to run multimedia application on a computer is:

(A) Gigabit card
(B) Network interface card
(C) Video graphic card
(D) Modem card

Q8. An interface or connection point through which peripheral devices connect to the computer is called:

(A) Expansion slots
(B) Port
(C) Modem
(D) System Bus

Q9. TCP and UDP protocols are used at the following layer:

(A) Session layer
(B) Application layer
(C) Network layer
(D) Transport layer

Q10. To setup a computer lab of 25 computers which class of IP address is suitable?

(A) Class A
(B) Class B
(C) Class C
(D) Class D

Q11. Which of the following satellite systems is located above Low Earth Orbit and Below Geostationary Earth Orbit?

(A) Geostationary Earth Orbit
(B) High Earth Orbit
(C) Medium Earth Orbit
(D) Low Earth Orbit

Q12. A Radio Frequency (RF) communication system consists of two components known as:

(A) Transmitter and receiver
(B) Transmitter and medium
(C) Transmitter and transceiver
(D) Receiver and medium

Q13. Which of the following database languages is uses to insert, update and delete data in database?

(A) DDL
(B) DML
(C) DCL
(D) DXL

Q14. The maximum number of an instance of one entity associated with each instance of related entity is called:

(A) Unary relationship
(B) Modality
(C) Cardinality
(D) Relational schema

Q15. Which type of query is used to extract data from tables based on specified criteria?

(A) Action query
(B) Select query
(C) Update query
(D) Insert query

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> Note: COMPUTER SCIENCE HSSC–I. Time allowed: 2:40 Hours. Total Marks Sections B and C: 60. NOTE: The Questions of sections B and C are to be answered on the separately provided answer book. Use supplementary answer sheet i.e. Sheet–B if required. Write your answers neatly and legibly.

Q. SECTION – B (Marks 36)

Q. 2 Answer any TWELVE parts. All parts carry equal marks. (12 x 3 = 36)

(i) Briefly explain any two types of system software with examples.

(ii) Write down any two differences between hardware and software with examples.

(iii) What is printer? Briefly explain Laser Printer.

(iv) Write down two advantages and one disadvantage of flash memory.

(v) Briefly explain the Sequential and Direct Access Memory.

(vi) Identify the role of Database Administrator (DBA).

(vii) Briefly describe CISC and RISC CPU architectures with one example each.

(viii) Write down the purpose of following ports: a) Fire wire port b) Serial Port c) USB Port

(ix) Write down any two differences between Synchronous and Asynchronous Transmission with the help of diagram.

(x) If a college wants to setup a computer lab, which topology is best and why? Support your answer with two reasons.

(xi) Write down the protocol and its purpose used on following layers of OSI Model. Sr. No. | Layers of OSI Model | Protocol Used | Function/Purpose of Protocol — i: Data Link Layer; ii: Network Layer; iii: Session Layer.

(xii) Write down one advantage and two disadvantages of wireless communication.

(xiii) What is Infrared (IR)? Write its importance in wireless communication.

(xiv) Observe the E-R diagram and answer the following questions: COUNTRY —has— CAPITAL. a) Identify the cardinalities b) Identify the modality c) Identify the degree of relationship

(xv) Transform the following E-R diagram to the Relational Schema. COLLEGE —has— PRINCIPAL. a) Transform Entities b) Transform Relationship

(xvi) Briefly describe the following types of Form Layouts: a) Tabular b) Columnar

Q. SECTION – C (Marks 24)

Note: Attempt any THREE questions. All questions carry equal marks. (3 x 8 = 24)

Q. 3 Describe the following Motherboard components: (2+2+2+2) i) BIOS ii) Ports iii) PCI Expansion Slot iv) Cooling system

Q. 4 a) Write down any three differences between static and dynamic RAM. (2+2) b) Compare the TCP suite with OSI model. (04)

Q. 5 a) What is the difference between data bus and address bus? (2+2) b) Describe instruction cycle with the help of diagram. (3+1)

Q. 6 a) A company has a number of employees. Each employee may be assigned to one or more projects. A project must have at least one employee assigned and may have several employees assigned. i) Draw an ER Diagram ii) Identify cardinalities iii) Identify modalities (2+1+1) b) Elaborate primary and Alternate key with the help of pictorial example. (2+2)

> Note: —1HA-I 2212 (HA)—